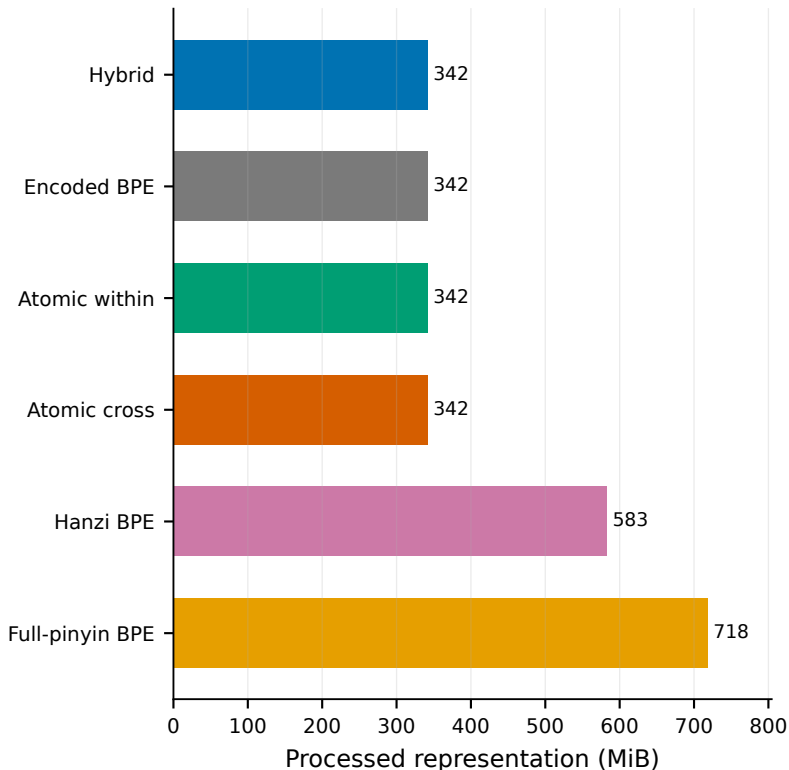


Representation and token efficiency on the same source documents

Hybrid Encoded BPE Atomic within Atomic cross Hanzi BPE Full-pinyin BPE

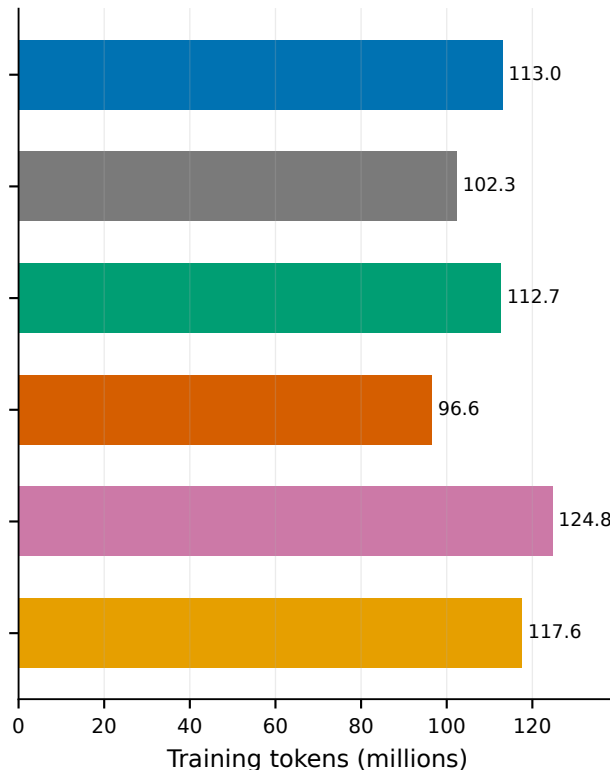
A

On-disk text



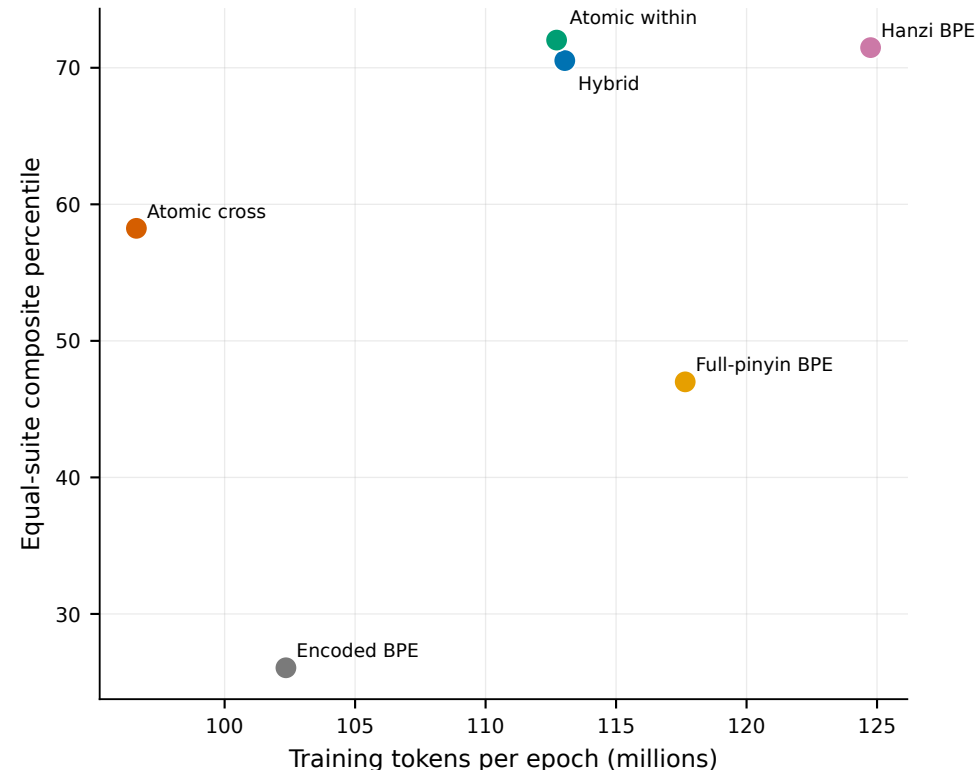
B

Tokenized sequence



C

Performance-token trade-off



All models see five epochs. UTF-8 text size is representation-dependent and is a storage measure, not corpus information content.